

Electric Vehicle Routing Problem With Variable Vehicle Speed and Soft Time Windows for Perishable Product Delivery

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Abstract—This work studies perishable products' distribution using electric commercial vehicles (ECVs). Extra energy is consumed for refrigeration to keep such products from deteriorating, which shortens the limited driving range of ECVs. Besides charging at public recharging stations, their travel speed can be adjusted to improve their driving range and decrease distribution cost. We propose an Electric Vehicle Routing problem with variable vehicle speed and soft time windows for perishable products (EVRP-VS). An energy consumption rate function of refrigerated ECVs during driving is introduced into the problem. The function considers refrigeration and has a nonlinear relationship with ECV speed and weight. As long as the carriage is not empty, the vehicle needs refrigeration during driving and during its stay at customers and stations. A mathematical programming model is developed for EVRP-VS, to minimize total distribution cost, including vehicle cost, power cost, refrigeration cost, and penalty cost due to delayed delivery. An adaptive hybrid ant colony optimization (AHACO) with a two-stage speed optimization strategy is proposed to solve EVRP-VS. In the first stage, local speed optimization is used to optimize the speed of each ant (ECV) in each transfer step. In the second one, global speed optimization further optimizes the speed for each fixed route constructed by ants. AHACO is applied to many problem instances. Experimental results show that it can effectively solve EVRP-VS in comparison with CPLEX and other meta-heuristics.

Index Terms—Electric commercial vehicle, vehicle routing problem, ant colony optimization, variable speed, perishable products.

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I. INTRODUCTION

PERISHABLE products, such as fruits, vegetables, seafood, and vaccines, may deteriorate during their transport processes. To keep them from deteriorating, it is necessary to transport and distribute them in a low-temperature environment. ECVs are one of the cleanest means of transport [1], which have no local Greenhouse Gas (GHG) emissions, and produce very low noise. With the development of ECV techniques, more and more ECVs are being deployed for last-mile deliveries in urban areas.

The driving range of ECVs is much shorter than that of fuel vehicles and not enough to perform the typical one-delivery tour [2], which is especially true for perishable products due to the extra energy consumption for refrigeration. ECVs can be recharged at public recharging stations along their routes [3], [4]. The energy consumption of ECVs heavily depends on the power required to accelerate and overcome rolling resistance, aerodynamic drag, grade resistance [5], and refrigeration. The aerodynamic drag depends on the speed of ECVs. Too high or low speed of a vehicle significantly increases the energy consumption and reduces the driving range. Moreover, the vehicle speed is required to be increased to fulfill customers' time window requirements and save refrigeration time. Therefore, it is important to optimize vehicle speed [6], and this is particularly practical for an autonomous vehicle logistic system (AVLS) consisting of ECVs without human drivers [7]. However, studies on an electric vehicle routing problem (EVRP) with variable speed are quite limited. There is only one paper on this topic. Xiao et al. [8] introduced an energy consumption rate (ECR) function of ECVs to EVRP for common cargo delivery and considered the traveling speed on each arc as a decision variable. The refrigeration required in perishable product delivery vehicles is not considered in their work. Customers have hard time windows and no delay is allowed. Public recharging stations are not considered either. For travel speed optimization, their ECR is considered as a piecewise linear function of speed and load of ECVs. Therefore, their model is simple and fails to sufficiently handle the practical issues.

This work proposes an Electric Vehicle Routing Problem with Variable vehicle Speed and soft time windows for perishable products (EVRP-VS). Different from existing EVRPs, a comprehensive ECR function of refrigerated ECVs during driving is newly introduced, which considers refrigeration and has a nonlinear relationship with their travel speed and a linear relationship with their weight. Furthermore, the energy

consumption of refrigeration is considered in the problem because perishable products must be kept at low temperatures. Besides driving, a vehicle needs refrigeration during its stay at customers (means at customer locations) and stations as long as its carriage is not empty. It stops cooling to conserve energy after visiting its last customer. Based on the ECR function, refrigeration cost, delay cost, and recharging station facilities, the traveling speed of a vehicle on each link of routes must be optimized, which is more difficult to solve than the existing speed optimization problems in various VRPs [15], [18], [28]. To our best knowledge, this is the first work considering the optimization of vehicle speed for EVRP with nonlinear ECR function.

We establish a mathematical programming model for EVRP-VS. The objective is to minimize the total distribution cost, including vehicle cost, power cost, refrigeration cost, and penalty cost for the delivery delay. An adaptive hybrid ant colony optimization approach (AHACO) is proposed to solve it. In AHACO, a two-stage speed optimization strategy is proposed to optimize vehicle speed. In the routing construction procedure of the first stage, when an ant selects its next arc to transfer, the traveling speed on the arc is optimized, which is called Local Speed Optimization. In the second stage, after a route is constructed by the ant, the speeds on all arcs in the route are jointly optimized, which is called Global Speed Optimization. Furthermore, an adaptive selection method is proposed to dynamically select local search operators according to their performance in previous iterations. AHACO is applied to many problem instances to verify its effectiveness.

We aim to make the below contributions: 1) we propose EVRP-VS that considers ECR, refrigeration, and soft time windows, such that vehicle speed is a decision variable to be optimized. ECR of refrigerated ECVs is a nonlinear function of their speed and vehicle weight. ECV needs refrigeration during driving and during its stay at customers and stations if it has non-empty carriage; we establish a mathematical programming model for EVRP-VS for the first time; and 2) we propose AHACO to solve EVRP-VS. The approach introduces a two-stage speed optimization strategy, multiple local search operators, and an adaptive selection method into Ant Colony Optimization (ACO) to tackle EVRP-VS. Experimental results and comparisons with CPLEX, Variable Neighborhood Search, and Adaptive Large Neighborhood Search show its outstanding solution ability.

The remainder of the paper is organized as follows: Session II briefly reviews the existing literature closely related to this work. In Session III, we introduce the notations in detail and provide mathematical programming formulations for EVRP-VS. Section IV describes AHACO. Experimental results are presented in Session V. Conclusions are drawn in Session VI.

II. LITERATURE REVIEW

A. Electric Vehicle Routing Problem

EVRP needs to consider charging during distribution. Felipe et al. [9] considered three charging strategies: fast, medium, and slow in their EVRPs. From the perspective of charging level, most of the literature assumes that ECVs must be fully

charged at recharging stations, while some literature allows partial recharging. Keskin et al. [10] showed that the partial recharging option might significantly improve the routing decisions. In terms of the relationship between charging quantity and charging time, it can be defined as linear functions or nonlinear functions [11].

Due to the limited battery capacity of ECVs, EVRP pays more attention to energy consumption. Most EVRPs assumed that the energy consumption of ECVs was linearly related to driving distance, but in fact, many factors affect the energy consumption of ECVs, such as speed, load, slope, and acceleration. Goeke et al. [3] presented a nonlinear energy consumption model that incorporates speed, slope, and cargo load distribution for the ECVs. The article focused on the impact of the load on the energy consumption of ECVs, but the travel speed was constant. Zhang et al. [12] considered the speed, slope, and other factors in the energy consumption of ECVs, but the speed was random and was not optimized. Basso et al. [13] introduced some coefficients of energy estimation for the road network into EVRP. Due to acceleration and braking, the speed of ECVs during driving is not constant and consists of three fixed speeds: starting, steady, and braking speed.

B. VRP With Variable Speed

Since the high carbon emission of fuel vehicles at low speed, some studies consider variable speed in Pollution Routing Problem (PRP) to reduce carbon emissions via changing the speed. Bektaş et al. [14] proposed PRP where the fuel consumption of a vehicle depended on its speed and load. Demir et al. [15] extended above PRP to consider hard time windows. They presented an Adaptive Large Neighborhood Search (ALNS) heuristic that combined a Speed Optimization Algorithm (SOA) based on their emissions function. SOA is used to obtain the optimal speed on each link of a fixed route generated by ALNS, so as to minimize the objective function comprising fuel consumption and driver wages. Jabali et al. [16] extended the PRP to Time-Dependent PRP without time windows, which considered the impact of traffic congestion on speed, allowing vehicles to wait at the depot or customers to avoid congestion. The speed is fixed during periods of traffic congestion and is optimized during the free-flow period. Franceschetti et al. [17] described a model for the Time-Dependent PRP with time windows. By considering time windows, the congestion period, and the speed range limitation, an algorithm was designed to optimize the departure time and speed on fixed routes. In summary, all the above Fuel VRPs with speed optimization are for the common cargo and do not consider time windows or only consider hard ones. As for the speed optimization method, most of the literature adopts the method of embedding SOA into meta-heuristics algorithms such as ALNS. In other words, the route is generated by a meta-heuristic firstly with a fixed speed, and then the traveling speed of each link on the fixed route is jointly optimized. However, the fixed routes for SOA only have the hard time windows constraint and the delay is not allowed.

Studies on EVRP with speed optimization are scarce. Xiao et al. [8] introduced an ECR into EVRP with hard time windows. ECR was simplified as a piecewise linear function of the speed. Therefore, their mathematics model is relatively simple and easy to solve.

In our study, ECVs are used for perishable product delivery. EVRP-VS considers the need for refrigeration, soft time windows and charging. Compared with existing EVRPs, EVRP-VS has the following unique characteristics. 1) A more comprehensive ECR function of a refrigerated ECV than existing ones is introduced, which considers refrigeration and is a nonlinear function of ECV's travel speed and weight. The use of non-linear ECR increases the complexity of EVRP-VS, leading to its nonlinear programming model; 2) during their delivery, ECVs must be refrigerated as long as there are products in their carriages. Hence, besides the power and refrigeration energy consumption during driving, when a vehicle stays at customers and stations, refrigeration is also required. The last customer visited by the vehicle must be identified in the mathematical model, which increases the difficulty of modeling. In addition, customers have soft time windows. A vehicle may arrive at customers outside their time windows, resulting in the refrigeration cost during its waiting at customers and penalty cost due to stockout.

In existing studies, the speed optimization of both fuel vehicles and ECVs only considers the power consumption during driving. In contrast, our study considers not only their power and refrigeration energy consumption during driving, but also the refrigeration energy consumption during their stay at customers and stations, as well as soft time windows when optimizing their traveling speed, which is more complex than existing speed optimization problems [15], [18], [28]. We develop a two-stage speed optimization approach embedded in each iteration of AHACO. In the first stage, the speed of each arc passed by the ant is optimized to minimize the transfer cost. Since the transfer cost is vital for the ant to choose its next node, the speed optimization can improve the routing decisions. In the second stage, after all the routes have been constructed by ants, a discretizing arrival time method which considers soft time window, power constraint, and recharging station nodes in the route is devised to jointly optimize the speeds on all links of each route to further optimize the route cost.

III. PROBLEM DESCRIPTION AND FORMULATION

A. EVRP-VS Model

Let $C = \{1, 2, \dots, n\}$ denote the set of customers, node 0 denote the depot, and node $n + 1$ denote the virtual depot. Every route starts from node 0 and ends at node $n + 1$. Let $F = \{n + 2, \dots, n + l\}$ denote the set of recharging stations which include virtual stations, and V' be the set of nodes with $V' = C \cup F$. Moreover, $V'_0 = V' \cup \{0\}$, $V'_{n+1} = V' \cup \{n + 1\}$, $V_0 = C \cup \{0\}$, and $V = V' \cup \{0\} \cup \{n + 1\}$.

EVRP-VS is described as a directed graph $G = (V, A)$ with the set of arcs $A = \{(i, j) | i, j \in V, i \neq j\}$. Each arc is associated with a distance d_{ij} (km) and a range of speed

$[v_{ij}^-, v_{ij}^+]$ (km/h). The speed of vehicles on the arc, v_{ij} , must be in the range of $[v_{ij}^-, v_{ij}^+]$. A fleet of ECVs is represented by $K = \{1, 2, \dots, m\}$. Each ECV has a net weight, W (kg), maximum battery capacity, P (kWh), and maximum load capacity, Q (kg). Each ECV is only allowed to deliver once. It leaves the depot with fully-charged batteries and finally returns to the depot. ECV can be charged once in a station during its route with a charging rate of w (kWh/h). The battery is always fully charged, i.e., the charge amount is $P - g_{ik}$, where g_{ik} is the remaining energy when the vehicle arrives at the station. As long as a vehicle is used, there is a fixed cost a (RMB per vehicle).

Each customer i in C must be served once with a single vehicle and is associated with a demand $q_i \leq Q$ (kg), and a time window $[e_i, l_i]$. If a vehicle arrives at customer i before e_i , service cannot be offered until time e_i . If the vehicle arrives at customer i after l_i , service is delayed, leading to a unit time penalty cost λ . Service duration time in customer i is s_i .

During the delivery period, perishable products need to be kept in a low-temperature environment. ECVs need refrigeration to meet this requirement, resulting in additional energy consumption. The needed temperature and difference between the temperature and external environment are known. For simplicity, assume that the cooling energy consumption per hour at this temperature difference is h (kWh/h). The refrigeration cost is mainly related to refrigeration time, including traveling, unloading, charging, and waiting time. Vehicles do not need refrigeration after visiting their last customers until they return to the depot.

The power of a vehicle also needs to consume electric energy. Every traveled arc consumes electric energy for both vehicle power and refrigeration. The energy consumption on the arc can be determined by the energy consumption rate f_{ij} (kWh/km) and the mileage of the arc, d_{ij} . f_{ij} is a nonlinear function of the speed and load of a vehicle. Let b (RMB/kWh) denote unit energy consumption cost. Then, the energy consumption costs of vehicle power and refrigeration on arc (i, j) are $(f_{ij} - h/v_{ij})d_{ij}$ and hd_{ij}/v_{ij} , respectively.

The decision variables of EVRP-VS are listed as follows.

Binary decision variables:

x_{ijk} : $x_{ijk} = 1$ if arc (i, j) is traversed by vehicle k ; 0, otherwise.

S_k : $S_k = 1$ if vehicle k is used; 0, otherwise.

L_{ik} : $L_{ik} = 1$ if vehicle k has unloaded all cargo on the vehicle at customer i ; 0, otherwise.

Non-negative continuous decision variables:

v_{ij} : traveling speed (km/h) on arc (i, j) .

t_{ij} : traveling time (hour) on arc (i, j) , $t_{ij} = d_{ij}/v_{ij}$.

y_{ik} : delay time if vehicle k arrives at customer i after l_i ; otherwise, $y_{ik} = 0$.

t_{ik}^-, t_{ik}^+ : arriving/departure time at node i for vehicle k , $t_{0k}^- = t_{0k}^+ = 0$.

u_{ik} : load of vehicle k when it arrives at node i .

g_{ik} : remaining energy of vehicle k when arriving at node i .

f_{ij} : electric energy consumption rate (kWh/km) of ECVs when traveling on arc (i, j) with a speed

of v_{ij} and a load of $u_{ik} - q_i$. It is defined in Equation (24).

Since the speed is continuous and the ECR is a nonlinear function, the EVRP-VS model is a nonlinear programming model. An effective method to linearize a model is discretizing speed within the speed range of each arc [14], [15].

Let v_{ijk}^r ($r = 1, 2, \dots, R$) denote R non-decreasing speed levels, Z_{ijk}^r is a binary variable that indicates whether or not arc (i, j) is traversed at speed level r by vehicle k . Let $W_{ijk} = \sum_{r=1}^R Z_{ijk}^r (\alpha (v_{ijk}^r)^2 + \beta v_{ijk}^r)$, $w_{ik} = \max\{e_i - t_{ik}^-, 0\}$. Then, the EVRP-VS with discretized speed is formulated as (1)–(22), shown on the next page. Objective (1) is to minimize the expected total cost. It contains four terms: 1) fixed cost related to each vehicle used; 2) refrigeration cost; 3) vehicle power cost; and penalty cost due to delay. Constraint (2) guarantees that all customers are only served exactly once. (3) establishes the continuity condition of the route. (4) and (5) ensure that each vehicle leaves the depot and returns to it, thereby limiting vehicle use to one trip. (6) guarantees that each visit to a recharging station is used at most once, i.e., not all recharging visit nodes must be used. Constraints (7)–(10) keep track of the battery state of charge in arc (i, j) when j is customer, recharging station, or depot, and ensure that it is not negative. (11) and (12) ensure that the sum of customer demands in a route does not exceed the vehicle capacity. (13) ensures that the traveling speed on arc (i, j) is within the actual driving speed limit of the arc. (14) and (15) represent the departure time of the vehicle from a customer and a recharging station, respectively. (16) and (17) ensure the time feasibility of arcs leaving the customers (or the depot) and the recharging stations, respectively, and M is an enough large positive number. (20)–(22) define the decision variables.

B. Energy Consumption Rate Model of ECVs

ECR represents the total amount of energy consumed per kilometer of the vehicle during driving. There has been no general method to calculate the ECR function for electronic vehicles (EVs) to date. The energy consumption of EVs is quite different from that of fuel vehicles. According to the electricity usage model in [19] and [20], the energy losses of an EV are distributed into four categories: Aerodynamic, Tires, Drivetrain, and Ancillary. Among them, the Aerodynamic losses are the major factor in the high-velocity regime, as the air resistance is proportional to the square of the speed.

Some literature reports have analyzed the road condition test data of various EV models and given the empirical curves of Speed-Energy Consumption. Straubel et al. [21] presented an empirical curve to show the relationship between the energy consumption and speed of Tesla Roadster. The curve is convex, indicating that the unit energy consumption per mile varies greatly as the speed increases.

As for ECVs, Goeke and Schneider [3] presented a realistic energy consumption model that incorporates speed, gradient, and vehicle mass. In the model, f denotes the ECR with a

load u , speed v , and the gradient angle α :

$$f = \Phi^d \varphi^r \left\{ \frac{1}{2} C_d \rho A v^2 + (W + u) (\sin \alpha + C_r \cos \alpha) \right\} \quad (23)$$

where, C_d is the aerodynamic drag coefficient, ρ is the air density, A is the frontal area of the vehicle, W is the vehicle curb weight, C_r is the rolling friction coefficient, and Φ^d and φ^r are the regression coefficients.

In the above model, as the speed approaches 0, ECR becomes smaller. This is inconsistent with the road test results of Tesla Roadster. The reason is that the model only considers Aerodynamics and Tires, but does not consider the energy consumption related to Drivetrain and Ancillary.

In summary, the above studies show that an ECR function is nonlinear and has a downward-canvas curve. The speed to minimize the ECR function must be greater than 0. Moreover, in our study, vehicles need refrigeration during driving, which also consumes electric energy. Therefore, we define the ECR function f of refrigerated ECVs as:

$$f = \alpha v^2 + \beta v + \theta (W + u) + \gamma + h/v \quad (24)$$

where coefficients α, θ , and γ are positive, and β is negative.

IV. ADAPTIVE HYBRID ACO FOR EVRP-VS

VRP is NP-hard, such that most solution approaches are heuristic-based. Ant Colony Optimization, first proposed by Dorigo et al. [25], is based on the behavior of ants foraging for food. ACO has been widely used in various VRPs and proved to be one of the most effective approaches for VRP, especially when combined with a specialized local search procedure [26], [27]. In this study, we propose an adaptive hybrid ACO (AHACO) for EVRP-VS with continuous speed. The framework of AHACO is given in Algorithm 1. In each iteration, the algorithm needs to perform the following four procedures.

First, in the *Route Construction* procedure, each ant in the ant colony constructs a feasible solution. When determining the next node to be visited by the ant, the probability of the ant transferring to each candidate node needs to be calculated based on factors such as pheromone concentration and transfer cost. Then, the ant randomly selects a candidate node to visit according to the above probability. The transfer cost to each candidate node depends on the speed, which directly affects the energy consumption cost caused by power and refrigeration to the candidate node, as well as the time delay cost. Therefore, we optimize the speed on the arc to each candidate node to obtain its minimum transfer cost and then apply it to the calculation of probability to construct more reasonable routes. Second, a *Local Search* procedure is used to improve the cost of each feasible solution generated by each elite ant. We develop six problem-specific local search operators. Due to different operators may have different abilities to improve the solutions, an adaptive selection method is proposed. The better the performance of the operator, the higher the probability to be selected, so as to give full play to

$$\min a \sum_{k \in K} \sum_{i \in C} x_{0ik} + bh \sum_{k \in K} \sum_{i \in C} (t_{ik}^+ - t_{0k}^+) L_{ik} + b \sum_{k \in K} \sum_{i \in V'_0} \sum_{j \in V'_{n+1}} \{W_{ijk} + \theta(W + u_{ik} - q_i) + \gamma\} d_{ij} x_{ijk} + \lambda \sum_{k \in K} \sum_{i \in C} y_{ik} \quad (1)$$

$$\text{s. t. } \sum_{i \in V'_0, i \neq j} \sum_{k \in K} x_{ijk} = 1, \forall j \in C \quad (2)$$

$$\sum_{i \in V'_0, i \neq j} x_{ijk} - \sum_{i \in V'_{n+1}, i \neq j} x_{jik} = 0, \forall j \in V', k \in K \quad (3)$$

$$\sum_{i \in V'} x_{0ik} = \sum_{j \in V'} x_{j,n+1,k} = S_k, \forall k \in K \quad (4)$$

$$S_k = \begin{cases} 1, & \text{if } \sum_{i \in V'_0} \sum_{j \in V'_{n+1}} x_{ijk} \geq 1, \forall k \in K \\ 0, & \text{else} \end{cases} \quad (5)$$

$$\sum_{i \in F, j \in V'_{n+1}} x_{ijk} \leq 1, \forall k \in K \quad (6)$$

$$g_{jk} \leq P - \left\{ W_{ijk} + \theta(W + u_{ik} - q_i) + \gamma + h \sum_{r=1}^R Z_{ijk}^r / v_{ijk}^r \right\} d_{ij} + M(1 - x_{ijk}), \forall i \in F \cup \{0\}, j \in V', k \in K \quad (7)$$

$$g_{jk} \leq g_{ik} - \{W_{ijk} + \theta(W + u_{ik} - q_i) + \gamma\} d_{ij} - h \left(s_i + d_{ij} \sum_{r=1}^R (Z_{ijk}^r / v_{ijk}^r) (1 - L_{ik}) + w_{ik} \right) + M(1 - x_{ijk}), \forall i \in C, j \in V', k \in K \quad (8)$$

$$g_{jk} \leq g_{ik} - \{W_{ijk} + \theta(W + u_{ik} - q_i) + \gamma\} d_{ij} - h(s_i + w_{ik}) + M(1 - x_{ijk}), \forall i \in C, j \in \{n+1\}, k \in K \quad (9)$$

$$g_{jk} \leq P - \{W_{ijk} + \theta(W + u_{ik} - q_i) + \gamma\} d_{ij} + M(1 - x_{ijk}), \forall i \in F, j \in \{n+1\}, k \in K \quad (10)$$

$$u_{0k} = \sum_{i \in V'_0} \sum_{j \in C} q_j x_{ijk} \leq Q, \forall k \in K \quad (11)$$

$$u_{jk} \leq u_{ik} - q_i x_{ijk} + Q(1 - x_{ijk}), \forall i \in V'_0, j \in V'_{n+1}, k \in K \quad (12)$$

$$v_{ij}^- \leq \sum_{r=1}^R Z_{ijk}^r v_{ijk}^r \leq v_{ij}^+, \forall i, j \in V, k \in K \quad (13)$$

$$t_{ik}^+ \geq t_{ik}^- + (P - g_{ik})/w, \forall i \in F, k \in K \quad (14)$$

$$t_{ik}^+ \geq \max\{e_i, t_{ik}^-\} + s_i, \forall i \in C, k \in K \quad (15)$$

$$\max\{e_i, t_{ik}^-\} + s_i + d_{ij} \sum_{r=1}^R Z_{ijk}^r / v_{ijk}^r - M(1 - x_{ijk}) \leq t_{jk}^-, \forall i \in V_0, j \in V'_{n+1}, k \in K \quad (16)$$

$$t_{ik}^- + \frac{P - g_{ik}}{w} + d_{ij} \sum_{r=1}^R Z_{ijk}^r / v_{ijk}^r - M(1 - x_{ijk}) \leq t_{jk}^-, \forall i \in F, j \in \{V'_{n+1} \setminus F\}, k \in K \quad (17)$$

$$L_{ik} = \begin{cases} 1, & \text{if } u_{ik} = q_i, \forall i \in C, k \in K \\ 0, & \text{else} \end{cases} \quad (18)$$

$$y_{ik} = \begin{cases} t_{ik}^- - l_i, & \text{if } l_i < t_{ik}^-, \forall i \in C, k \in K \\ 0, & \text{else} \end{cases} \quad (19)$$

$$\sum_{r=1}^R Z_{ijk}^r = x_{ijk}, \forall i \in V'_0, \forall j \in V'_{n+1}, i \neq j, \forall k \in K \quad (20)$$

$$Z_{ijk}^r \in \{0, 1\}, \forall i \in V'_0, j \in V'_{n+1}, k \in K, r = 1, \dots, R \quad (21)$$

$$x_{ijk} \in \{0, 1\}, v_{ij} > 0, \forall i \in V'_0, j \in V'_{n+1}, k \in K \quad (22)$$

the role of those operators with superior performance. Third, after each elite solution is improved by local search, we further optimize the speed on each arc of the solution to reduce the cost of the solution. Although the speed of each ant's transfer is optimized in the *Route Construction* procedure, the speed on each arc is not necessarily optimal in terms of the whole route. Therefore, it is necessary to jointly optimize the speeds on the arcs of each route in the solutions, to further minimize the cost of each solution. Finally, the pheromone concentration on each arc in the graph is updated by using the above feasible solutions. The algorithm runs until a maximum number of iterations $\hat{L}$ is reached and returns the best-so-far solution s^* .

Algorithm 1 AHACO

Initialization**Repeat**

For each ant **do**//each ant construct a feasible solution

Route Construction Procedure

End For

Local Search Procedure

Global Speed Optimization Procedure

Pheromone Updating Procedure

Until termination criterion is met

Return s^*

AHACO is different from existing ACOs in the following aspects: 1) the speed of each ant is optimized when it transfers each time, 2) the speeds on all arcs in the route are further jointly optimized after constructing the routes, and 3) the local search operators are selected by using an adaptive selection method.

A. Route Construction

In AHACO, an ant represents a vehicle and starts from the depot at the initial time. Restricted by the vehicle capacity and battery capacity, it can only visit some customers and then return to the depot to construct a tour (route). During the above process, it can be charged in a station when the power is not enough. After the ant returns to the depot, it represents another vehicle and starts from the depot again to visit the remaining customers to construct another route. This process is repeated until all customers are visited, and a solution is composed of multiple routes constructed by the ant.

1) *Candidate Node Set*: When the ant is in node i , it needs to choose the next node j . As we know, nodes i and j have three types: depot, customers, and stations. We need to determine the candidate node set G_k for ant k to visit. The G_k consists of all nodes that meet the vehicle capacity constraint, i.e., $u_{ik} + q_j \leq Q$, and power constraints. The power constraints need to ensure that ant k has enough residual power to return to the depot or the nearest station after arriving at node j .

2) *Transition Probability*: The ant randomly selects a node j from G_k as its next node according to the following probability.

$$\rho_{ij}^k = \begin{cases} \frac{[\eta_{ij}]^{\sigma_1} [\tau_{ij}]^{\sigma_2} [\mu_{ij}]^{\sigma_3}}{\sum_{h \in G_k} [\eta_{ih}]^{\sigma_1} [\tau_{ih}]^{\sigma_2} [\mu_{ih}]^{\sigma_3}}, & \text{if } j \in G_k \\ 0, & \text{else} \end{cases} \quad (25)$$

where ρ_{ij}^k is the probability of choosing node j as the next node of i . It is determined by the cost, savings, and pheromone accumulation. η_{ij} is defined as the reciprocal of the transfer cost from node i to j . τ_{ij} denotes the amount of pheromone on the path between the current node i and node j . $\mu_{ij} = d_{0i} + d_{0j} - d_{ij}$, denotes the saving cost for combining customer i and j into one tour. σ_1, σ_2 , and σ_3 are weight coefficients.

3) *Local Speed Optimization*: Equation (25) shows that transfer cost is a key issue to impact the ants' transfer decisions. Therefore, to improve routing decisions, it is necessary to determine the optimal speed in the speed range of arc (i, j) to minimize transfer cost. Since speed optimization is only for one arc, it is called Local Speed Optimization. Let C_{ij} denote the transfer cost in arc (i, j) .

$$\begin{aligned} C_{ij}(v_{ij}) = & bd_{ij} (\alpha v_{ij}^2 + \beta v_{ij} + \theta (W + u_i - q_i) + \gamma) \\ & + bh (d_{ij}/v_{ij} + \max \{e_j - t_i^+ - d_{ij}/v_{ij}, 0\}) \\ & + \lambda \max \{t_i^+ + d_{ij}/v_{ij} - l_j, 0\} \end{aligned} \quad (26)$$

where the first term is power cost on arc (i, j) , the second one is refrigeration cost during the driving on arc (i, j) and waiting at node j , and the third one is time delay cost at node j . Note that when j is a recharging station or depot, there is no time delay cost. $v_{ij} \in [v_{ij}^-, v_{ij}^+]$ is a decision variable. Since $\eta_{ij} = 1/C_{ij}$, to obtain η_{ij} in (25), we need to find the minimum transfer cost, C_{ij}^* , and the corresponding optimal speed, v_{ij}^* .

An ECV runs at a certain speed within the range of $[v_{ij}^-, v_{ij}^+]$ on arc (i, j) . The interval of its arrival time at node j is $[\tilde{T}_j^-, \tilde{T}_j^+]$. Compared with $[e_j, l_j]$, there are six cases in this time interval: 1) $t_i^+ + d_{ij}/v_{ij}^- < e_j$; 2) $t_i^+ + d_{ij}/v_{ij}^+ > l_j$; 3) $e_j \leq t_i^+ + d_{ij}/v_{ij}^- < t_i^+ + d_{ij}/v_{ij}^+ \leq l_j$; 4) $t_i^+ + d_{ij}/v_{ij}^- < e_j < t_i^+ + d_{ij}/v_{ij}^+ < l_j$; 5) $e_j < t_i^+ + d_{ij}/v_{ij}^- < l_j < t_i^+ + d_{ij}/v_{ij}^+$; and 6) $t_i^+ + d_{ij}/v_{ij}^+ < e_j < l_j < t_i^+ + d_{ij}/v_{ij}^-$. We can obtain the optimal speed for each case.

B. Local Search

This procedure attempts to further decrease the total cost of a feasible solution by iteratively moving to a neighbor solution. We adopt three LS operators in [23], and design other three new LS operators for EVRP-VS. These operators are used to improve each solution obtained by elite ants. Moreover, since different operators may have different abilities to improve the solutions, we propose an adaptive selection method to select operators according to their previous performance.

1) *Local Search Operators*: (1) Node Exchange. We adopt three Node Exchange operators in [23]: Exchange (ES for short), 2-opt operators of Sing-route, and Exchange operator of Multi-route (EM).

(2) Maximum Delay (MD). When a vehicle arrives at a customer after the time window, the service is delayed and a time penalty cost is incurred. Hence, reducing such cost contributes to the reduction of the total distribution cost. In the MD operator, we select the customer with the largest delay time in the solution and remove it, and then reinsert it into the appropriate location to minimize the total distribution cost.

(3) Maximum Speed-Difference (MS). During the transfer process, the optimal speed v'_{ij} is calculated to minimize the transfer cost C_{ij} . However, the actual optimal speed v^*_{ij} may not equal v'_{ij} due to the actual speed limit of the arc. In the MS operator, we select arc (i, j) with the maximum difference between v'_{ij} and v^*_{ij} in a solution, i.e., $\max\{|v'_{ij} - v^*_{ij}|\}$, remove node j from the route, and then reinsert it into the appropriate position to further reduce the distribution cost.

(4) Worst Recharging Station (WS). The motivation of this operator is to use the battery as much as possible before charging is needed. First, the station that an ECV visits with the highest battery level is removed. Then, we determine the best station insertion between the customers that the ECV arrives with a negative battery level and the removed station. We select the best feasible insertion and perform it.

2) *Adaptive Selection of Operators*: When the LS operators are combined with ACO, Serial Selection mode (SSM) is usually used to select LS operators. The SSM is to select operators one by one according to a certain order, and each operator is executed for a certain number of iterations.

In this paper, we propose another mode: Adaptive Selection mode (ASM). In this mode, operators are selected dynamically and adaptively based on a probability calculated by using their performances in previous selections (calls). To calculate the selection probabilities, an adaptive weight w and a score π are assigned to each operator. The better the performance of the operator, the higher score and weight of the operator, and the higher the probability of being selected.

In a call, if a new best solution has been found, then the score corresponding to the operator who achieved that solution are increased by θ_1 . If the operator has yielded a better solution than the previous, while being worse than the best solution so far, then the score is increased by θ_2 . If the new solution is worse than the previous one but accepted by using the Simulated Annealing rule, then the score is increased by θ_3 . Otherwise, the score is increased by 0.

In each iteration of AHACO, the *Local Search procedure* includes $\bar{L}$ calls for each elite solution. At the end of call t , the weight of operator d is updated using the formula:

$$\omega_d^{t+1} = \begin{cases} (1 - r_p) \omega_d^t + r_p \frac{\pi_d^t}{\mu_d^t}, & \mu_d^t > 0 \\ \omega_d^t, & \mu_d^t = 0 \end{cases} \quad (27)$$

where $r_p \in (0, 1)$ is the roulette wheel parameter, μ_d^t is the number of times that operator d was used up to call t , and π_d^t is the score associated with operator d in call t . After

updating the weights, the probabilities of the operators which will be used in the next call ($t+1$) are calculated by $\rho_d^{t+1} = \omega_d^{t+1} / \sum_{d=1}^6 \omega_d^{t+1}$.

3) *Acceptance Criteria*: To improve the global search capability, the Simulated Annealing is used to accept or reject a solution. If the number of vehicles in the new solution (S_{new}) is smaller than that of the best solution found so far, or if the two solutions have the same number of vehicles but S_{new} has a smaller total cost, then S_{new} is accepted to replace the current solution (S_c). S_{new} is rejected if it needs more vehicles. If the numbers of vehicles are equal but S_{new} has a larger cost, S_{new} is accepted with probability $P_L = e^{-(c(S_{\text{new}}) - c(S_c))/T}$, where $c(X)$ is the total cost of solution X , T is the current temperature. T is initially set to $T_0 = \eta c(S_{\text{elite}})$ and decreased at each call using the formula $T = \delta T$, where S_{elite} is the elite solution for the local optimization, η is the initial temperature parameter, and δ ($0 < \delta < 1$) is the cooling rate parameter.

C. Global Speed Optimization

When an ant makes the transfer decision, it chooses the best speed in the actual speed range of the arc to minimize the transfer cost, so as to construct a more reasonable route. However, this local speed optimization only focuses on the current arc, without considering the impact on subsequent nodes. Due to the speed limit and the soft time window, it may lead to uneconomical driving on the subsequent arc even if driving at the optimal speed on the current arc. For instance, as the driving speed of the current arc is too slow, the next arc will arrive late even at the maximum speed. Therefore, it is necessary to jointly optimize the speeds on all arcs in each route, to make the whole route have the lowest cost. Since the speed optimization is for all arcs of a route, it is called Global Speed Optimization.

We select the solutions generated by the elite ants to optimize the speeds on all arcs of each route. The Global Speed Optimization Problem for a route with an ECV (GSOP-EV) is defined on a feasible route $(0, 1, \dots, r, n+1)$ of nodes (may include a recharging station), which are all served by a single ECV, where nodes 0 and $n+1$ denote the depot.

Like EVRP-VS, GSOP-EV is also a non-linear programming problem, which can be solved by non-linear programming software or some meta-heuristics such as Genetic Algorithm. However, even for a small-size instance (about 10 nodes), the above methods would be very time-consuming due to the large number of sub-problems needed to be solved in AHACO. Instead, we propose a rapid algorithm for GSOP-EV.

The algorithm is based on the Discretizing Arrival Times (DAT) method developed to solve the Speed Optimization Problem (SOP) in [18] and [28]. The basic idea of DAT is that the arrival time of each node is discretized, and the problem can be regarded as a Shortest Path Problem on an acyclic graph. However, it cannot solve GSOP-EV. The SOP in [18] and [28] solved by DAT is simple. There is only a hard time window constraint and the delay is not allowed. GSOP-EV is much more complex than the SOP. Apart from a soft time window constraint, we have power constraints and a station node in the route. Hence, we developed a

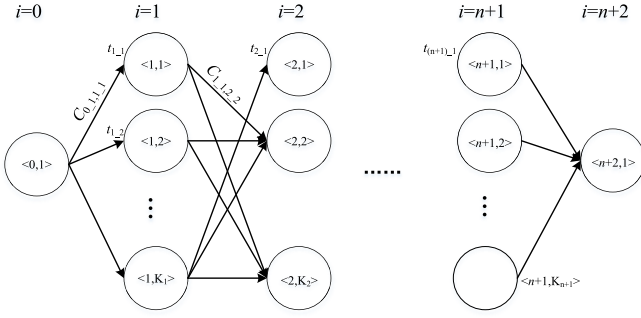


Fig. 1. Graph obtained by discretizing arrival times of GSOP-EV.

DAT-based algorithm for GSOP-EV. The algorithm consists of three steps:

First, the earliest and latest arrival time of each node along the route is calculated by ignoring power constraints and then discretized. Let $\check{t}_{i,i+1} = d_{i,i+1}/v_{i,i+1}^+$ and $\hat{t}_{i,i+1} = d_{i,i+1}/v_{i,i+1}^-$ represent the minimum and maximum travel time in arc $(i, i+1)$ respectively. By considering the earliest service starting time (e_i) for customers, starting from $i = 1$, we calculate $\check{T}_i$ and $\hat{T}_i$ of each node along the route via $\check{T}_i = \max \{ \check{T}_{i-1}, e_{i-1} \} + s_{i-1} + \check{t}_{i-1,i}$, and $\hat{T}_i = \max \{ \hat{T}_{i-1}, e_{i-1} \} + s_{i-1} + \hat{t}_{i-1,i}$. If node $(i-1)$ is a station, s_{i-1} is replaced by P/w . Then, we discretize the arrival time in $[\check{T}_i, \hat{T}_i]$ for each node i . Let K_i denote the arrival time discretization level for node i . Node i in the route (except node 0) is replicated into K_i nodes. Each of these new replicated nodes is represented as vertex $\langle i, p \rangle$, where i is the index of the node in the route, and p refers to the discretized arrival time $t_{i,p}$ at the node.

Second, we connect every pair of vertexes whose corresponding nodes are adjacent in the route and construct a directed graph as shown in Fig. 1. Let arc $(\langle i, p \rangle, \langle i+1, p' \rangle)$ represent the link between vertexes $\langle i, p \rangle$ and $\langle i+1, p' \rangle$. As for arc $(\langle i, p \rangle, \langle i+1, p' \rangle)$, since $t_{i,p}, t_{(i+1),p'}, s_i$, and e_i are known, we can calculate $t_{i,p,(i+1),p'}$ and $v_{i,p,(i+1),p'}$ that represent the time and speed at which a vehicle travels from vertexes $\langle i, p \rangle$ to $\langle i+1, p' \rangle$. The cost in each arc is calculated. Let $C_{i,p,(i+1),p'}$ denote the cost in arc $(\langle i, p \rangle, \langle i+1, p' \rangle)$, which contains power and refrigeration cost during driving, and time delay cost and refrigeration one at vertex $\langle i, p \rangle$.

$$C_{i,p,(i+1),p'} = b(\alpha (v_{i,p,(i+1),p'})^2 + \beta v_{i,p,(i+1),p'}) + \theta (W + u_i - q_i) + \gamma d_{i,i+1} + bh(t_{(i+1),p'} - t_{i,p}) + \lambda \max \{ t_{i,p} - l_i, 0 \} \quad (28)$$

To make the graph have only one terminal node, a virtual node $(n+2)$ is added after node $(n+1)$, which is connected with all arrival time vertexes of node $n+1$ without any cost.

Finally, considering the power constraint, if a path through this graph, from node 0 to node $n+2$, meets the power constraint, it is a feasible solution of GSOP-EV. Finding the shortest path through the network can be regarded as finding the optimal arrival time for each node. Therefore, this is a constrained shortest path problem (CSP), and the branch and

bound algorithm is applied to solve it. Given the optimal arrival time t_i^* and t_{i+1}^* of nodes i and $i+1$, the optimal speed $v_{i,i+1}^*$ on arc $(i, i+1)$ is determined by $v_{i,i+1}^* = d_{i,i+1}/(t_{i+1}^* - (\max \{ t_i^*, e_i \} + s_i))$. Thus, we can obtain the optimal speed for each arc of the route.

The minimum cost of the CSP is compared with the original cost of the route. If there is an improvement, use the new solution to replace the original one.

D. Pheromone Updating

Pheromone updating is conducted by firstly reducing the amount of pheromone on all routes to simulate the evaporation. Afterward, ants lay pheromone on the paths they traversed. To speed up the convergence of the algorithm, we use the elite ant strategy, i.e., the N best ants of the iteration are allowed to lay pheromone twice on the paths they traversed [25]. All solutions constructed by ants in this iteration are sorted from best to worst. All of these are done via:

$$\tau_{ij}^{new} = \rho \tau_{ij}^{old} + \sum_{k=1}^M \Delta \tau_{ij}^k + \sum_{k=1}^N \Delta \tau_{ij}^k \quad (29)$$

where τ_{ij}^{old} and τ_{ij}^{new} are the pheromone on arc (i, j) before and after updating, ρ is the constant that controls the speed of evaporation, $\Delta \tau_{ij}^k$ is the increased pheromone on arc (i, j) of solution k . If arc (i, j) is on solution k , then $\Delta \tau_{ij}^k = 1/c(S_k)$; otherwise, $\Delta \tau_{ij}^k = 0$. Moreover, to avoid local optima and increase the probability of finding high-quality solutions, the amount of pheromone in each arc is limited in a range $[\tau_{min}, \tau_{max}]$ [29]. $\tau_{max} = 1/\{(1-\rho)c(S_{IS})\}$, $\tau_{min} = (1-\rho)/c(S_{IS})$, where S_{IS} is the cost of an initial solution. The pheromone trails are initialized with τ_{max} .

V. EXPERIMENTAL RESULTS

AHACO is applied to 24 small-scale instances and compared with CPLEX. Then, three ACOs are used to solve large-scale instances to compare the performance of two operator selection modes and show the role of each LS operator in AHACO. Two experiments are conducted to verify the effect of our speed optimization approach. The first one is conducted to verify the effect of a DAT-based method for GSOP-EV, and the second one is to compare Local Speed Optimization with Global one and verify the two-stage speed optimization strategy. Finally, AHACO is compared with other two meta-heuristics. All algorithms are coded in C++ and run on a PC with an Intel Core i7-6600U CPU and 16 GB RAM.

A. Problem Instances

Two sets of benchmark instances for EVRP with time windows in [2] are used. The first one has 30 large-scale ones, each of which contains 100 customers and 21 stations. The second one has 24 small-scale ones which include two subsets of 12 problems, each involving 5 and 10 customers drawn randomly from the large instances. Each set includes three classes where customers and stations are clustered (c), randomly distributed (r), and both clustered and randomly distributed (rc).

We set: $v_{ij}^- = 30$ km/h, $v_{ij}^+ = 60$ km/h, $h = 4$ kWh/h, $a = 100$ RMB per vehicle, $\lambda = 600$ RMB/h, $b = 1$ RMB/kWh, $\alpha = 0.00002771$, $\beta = -0.003$, $\theta = 0.0001$, and $\gamma = 0.197$. Other parameters are same as those in [2].

B. Algorithm Parameters

AHACO has 14 parameters falling into three groups: ACO, local search, and Simulated Annealing ones. Empirical observations in some experiments show that 50 ants with 5 elite ants (10 ants with 1 elite ant), $\hat{L} = 100$, and $\bar{L} = 50$ are enough for AHACO to obtain a high-quality solution for large-scale (small-scale) instances. Hence, we adopt those setting. Moreover, we set $\rho = 0.9$ as in [12] and [26], and $\delta = 0.99975$ as in [17]. Other eight parameters are determined by the parameter-tuning experiments conducted on six large-scale instances (c101_21, c201_21, r101_21, r201_21, rc101_21, and rc201_21). The final parameter values are set as $\sigma_1 = 2$, $\sigma_2 = 1$, $\sigma_3 = 2$, $\theta_1 = 5$, $\theta_2 = 3$, $\theta_3 = 1$, $r_p = 0.2$, and $\eta = 0.1$.

C. Small-Scale Example Experiments

The small-scale instance set is used to compare AHACO with CPLEX. The runtime of CPLEX is restricted as 10800 seconds for all instances. Since the EVRP-VS model with continuous speed is nonlinear and nonconvex, CPLEX cannot ensure finding the optimal solution. However, it can find the optimal solution to EVRP-VS with discretized speed as the model is linear. The speed discretization level R is set to be 30 in this experiment.

AHACO performs 10 runs for each instance of the EVRP-VS model with continuous speed. Table I presents the results. Column ‘Best’ and ‘Avg’ indicate the best and average solution(s) in 10 runs respectively, ‘time(s)’ indicates the computing time of the AHACO or CPLEX in seconds, ‘Gap’ means the percentage deviations of the solution obtained with CPLEX from the best solution found by AHACO. ‘-’ means no solution is found by CPLEX.

The results show that the average gap between the solution obtained with CPLEX and the best solution produced by AHACO is as high as 452.78%. The reasons are twofold: first, CPLEX solves EVRP-VS with discretized speed $R = 30$, while AHACO does so with continuous speed. As we know, the feasible solution set of the former is a subset of the latter, so the optimal cost of the latter is often smaller than that of the latter. Second, due to the complexity of EVRP-VS with discretized speed, CPLEX under given runtime obtains the optimal solution for 10 instances, but for the other 14 instances, only the upper bound or no solution (c101C10) can be found.

D. Large-Scale Example Experiments

We first compare the performance of the two operator selection modes (SSM and ASM), and then show the role of each LS operator in AHACO under the influence of ASM.

TABLE I
RESULTS OF SMALL INSTANCES

Instance	AHACO			CPLEX($R=30$)		
	Best	Avg	time(s)	Value	time(s)	Gap%
c101C5	167.03	167.03	2.69	167.83	2620	0.48
c103C5	171.85	171.85	2.6	173.51	31.75	0.96
c206C5	172.69	172.69	2.68	173.46	10800	0.45
c208C5	160.44	160.44	2.49	161.35	4571.58	0.57
r104C5	126.89	126.89	2.78	127.63	2531.34	0.58
r105C5	131.74	131.74	2.84	134.35	553.97	1.98
r202C5	141.6	141.6	2.69	142.13	851.67	0.37
r203C5	155.64	155.64	2.69	156.10	1857.64	0.30
rc105C5	157.45	157.45	2.77	226.60	10800	43.92
rc108C5	154.38	154.38	2.81	155.16	5545.16	0.50
rc204C5	164.66	164.66	2.76	166.37	4082.73	1.04
rc208C5	144.44	144.44	2.63	145.21	1996.42	0.53
c101C10	224.46	224.46	9.13	-	-	-
c104C10	150.66	150.66	5.61	638.21	10800	323.61
c202C10	195.07	195.07	9.29	4331.36	10800	2120.41
c205C10	183.57	183.57	8.07	924.58	10800	403.67
r102C10	217.58	217.58	6.76	992.83	10800	356.30
r103C10	134.46	134.46	6.61	231.21	10800	71.95
r201C10	170.17	170.23	8.09	3590.65	10800	2010.04
r203C10	163.62	163.62	11.21	725.83	10800	343.61
rc102C10	288.71	288.71	7.12	2696.22	10800	833.88
rc108C10	218.09	218.09	6.44	3078.49	10800	1311.57
rc201C10	200.27	200.56	8.65	2177.02	10800	987.04
rc205C10	185.65	187.08	8.4	3156.36	10800	1600.17
Avg.	174.21	174.29	5.33	1064.02	7175.75	452.78

1) *Comparison of SSM and ASM*: The ASM method in AHACO is replaced by SSM to form a heuristic termed SHACO. Then, ACO, SHACO, and AHACO are applied to the 30 large-scale instances of EVRP-VS with continuous speed. ACO is AHACO without the local search and global speed optimization. Each algorithm runs 10 times. The results are shown in Fig. 2. The two columns in the Figure represent the average cost of the best solution and the average solution of 30 instances, respectively. The dotted line denotes the average computing time of the best solution of 30 instances in seconds.

The results show that: 1) the best solutions found by the two hybrid ACOs are significantly better than ACO, with an average improvement of 22.45% and 14.29% respectively; 2) among the two modes, ASM is much better than SSM. Among 30 instances, the best solutions of all instances are achieved by AHACO; 3) ACO takes the least time, with an average of 42.72 seconds. Due to the local search and global speed optimization procedures, the time of the two hybrid ACOs increases greatly. AHACO takes an average of 161.91 seconds, which is longer than that of SHACO. In sum, AHACO can dynamically adjust the selection probability of each operator and randomly select operators according to the probability, and therefore, AHACO has the best performance.

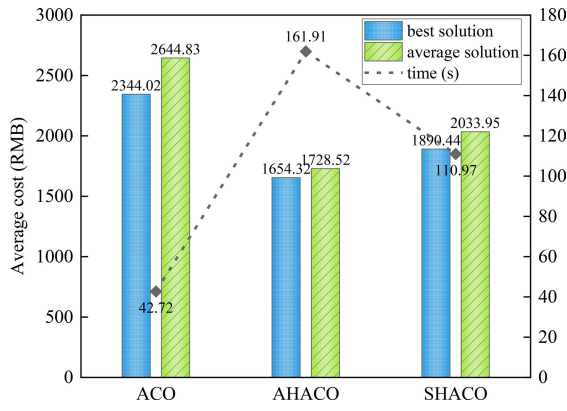


Fig. 2. Comparison of SSM and ASM.

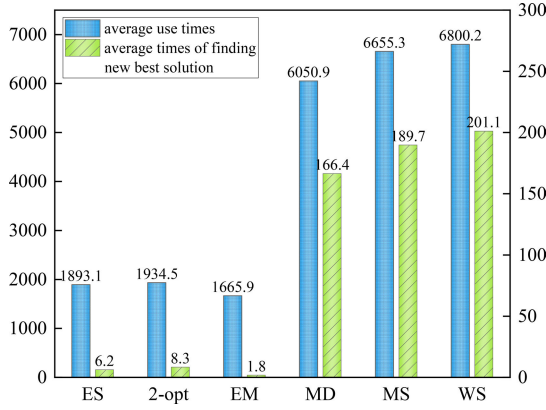


Fig. 3. Role of LS operators in AHACO.

2) *Performance of Each LS Operator in AHACO*: To verify the performance of each LS operator in AHACO, for each instance, we take the best solution and count the total number of times each operator is selected and the number of times each operator finds a new best solution. The results are shown in Fig. 3. The two columns in the Figure represent the average times of each operator being used and the average times of finding the new best solution by each operator for 30 instances, respectively.

It can be seen from Fig. 3 that there is a huge performance difference among the six LS operators, in terms of the number of average uses and average times of finding a new best solution. The average use times of ES, 2-opt, and EM is less than 2000, which is less than 1/3 of that of MD, MS, and WS. The reason may be that ES, 2-opt, and EM are general operators for VRPs, while MD, MS, and WS are specific for EVRP-VS. In terms of the times of finding a new best solution, ES and 2-opt have an average of only 6.2 and 8.3 times respectively, and EM that exchanges two nodes between two routes has only 1.8 times on average and can hardly find a new best solution. In sharp contrast to them, MD, MS, and WS have as high as 166.4, 189.7, and 201.1 times of finding new best solution respectively. This shows that the ability of MD, MS, and WS to improve the solution of EVRP-VS is much higher than that of ES, 2-opt, and EM.

E. Experiments on Speed Optimization

1) *Experiments for GSOP-EV*: The purpose of the experiments is twofold. Since a large number of GSOP-EVs are

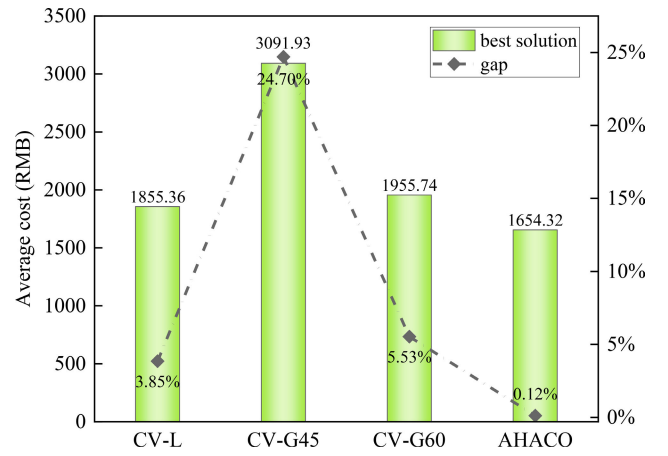


Fig. 4. Speed optimization in AHACO.

solved by the DAT-based algorithm in each iteration of AHACO, we aim to: 1) check whether the computational time of the method is small enough to meet the requirement of AHACO; and 2) verify the effect of Global Speed Optimization.

Two types of routes are used. First, we randomly select some routes constructed by ants during the process of AHACO for large-scale instances. These routes with locally optimized speed belong to the first type denoted by VSP. Then, the speed on each arc of these routes is replaced by a fixed speed (40 km/h) to generate routes of the second type (the speed can meet the power constraint of each arc), which is represented by FSP. In addition, computational time and optimization effect may be different due to different route lengths (nodes in the route). Therefore, we establish multiple instance sets with different route lengths and different route types for GSOP-EV. Route lengths are taken as 5, 10, 15, 25, and 45 respectively, and each set contains 100 instances (routes).

Arrival time discretization level affects the computational time and optimization effect of our DAT. The higher the discretization level, the smaller the speed interval, the more vertexes in the corresponding directed graph, and the higher computational time of the branch and bound algorithm. In our experiments, it is set as 5, 15, 30, 60, and 150.

DAT with different discretization levels is used to solve the instances in each set. It is noted that, for a total of 100 routes in each set, DAT may improve only a few routes. Statistics are made according to the instance set, including average computing time in milliseconds (t), the number of improved instances ($\hat{N}$), and the average improvement percentage of those improved instances ($\bar{R}$). The results are shown in Table II. We conclude that:

1) Applying Global Speed Optimization can improve the solutions for most of the FSP instances, but only a small part of the VSP instances. Both $\hat{N}$ and $\bar{R}$ of VSP instances are much lower than those of FSP instances.

2) For FSP instances, most instances are improved by DAT. $\hat{N}$ ranges from 51 to 97, and $\bar{R}$ is 7.39%~13.82%. The longer the route length and the higher the discretization level, the higher both $\hat{N}$ and $\bar{R}$. We have also found that the more nodes in the route, the more likely the customers are delayed. After

TABLE II
RESULTS OF GSOP-EV BY DAT METHODS WITH DIFFERENT DISCRETIZATION LEVELS

	DAT5			DAT15			DAT30			DAT60			DAT150		
	$\bar{R}\%$	$\hat{N}$	$t(\text{ms})$	$\bar{R}\%$	$\hat{N}$	$t(\text{ms})$	$\bar{R}\%$	$\hat{N}$	$t(\text{ms})$	$\bar{R}\%$	$\hat{N}$	$t(\text{ms})$	$\bar{R}\%$	$\hat{N}$	$t(\text{ms})$
VS5	0	0	6.65	0	0	10.21	0	0	12.41	0	0	18.27	0	0	91.01
VS10	1.44	22	9.99	1.57	23	13.31	1.61	23	18.76	1.54	24	30.50	1.55	24	118.44
VS15	2.28	15	18.83	2.29	15	28.45	2.30	15	36.44	2.30	15	57.81	2.32	15	223.35
VS25	1	14	27.52	1.02	15	39.01	1.03	17	53.72	1.03	17	146.57	1.04	17	901.26
VS45	0.71	18	34.74	0.71	18	47.79	0.73	19	127.40	0.74	19	351.68	0.76	19	2772.83
FS5	7.39	51	3.96	7.44	52	7.82	7.57	55	9.44	7.69	57	15.83	7.76	57	87.02
FS10	8.45	57	6.61	8.73	60	9.98	9.13	63	14.38	10.09	63	28.02	11.44	66	102.63
FS15	10.78	58	19.27	11.04	62	31.74	11.13	64	37.39	11.58	66	56.90	11.86	67	240.29
FS25	11.39	66	22.45	11.54	69	34.72	11.88	70	58.90	12.59	71	159.07	12.90	74	870.16
FS45	12.36	92	37.41	12.40	93	45.09	12.75	96	119.64	13.21	96	334.26	13.82	97	2219.62

applying Global Speed Optimization, the delay duration can be effectively reduced, so as to greatly reduce the cost.

3) For VSP instances, only a few instances are improved. $\hat{N}$ ranges from 0 to 24. $\bar{R}$ ranges from 0% to 2.32%, much lower than that of FSP instances. The reason is that for VSP instances, since Local Speed Optimization has been carried out for each arc on the routes, it is not easy for Global Speed Optimization to further improve the solutions.

4) The computing time of DAT for example sets is in the range of [3, 3000] ms. The more nodes in the route and the higher the discretization level, the more the time. Considering that both $\hat{N}$ and $\bar{R}$ of DAT60 differ little from DAT150, and its time is acceptable, DAT60 is adopted for Global Speed Optimization in AHACO.

2) Experiments on Two-Stage Speed Optimization:

In [8], [15], and [17], the fixed speed is used to generate the initial solution firstly, and then the Global Speed Optimization is carried out for the speed of each link on each route of the solution. Different from the approaches in the above literature, AHACO adopts the two-stage speed optimization strategy combining Local Speed Optimization and Global one. There have been no studies on Local Speed Optimization to date. We thus compare with Global Speed Optimization and verify the two-stage speed optimization strategy.

We use CV-L to denote AHACO with Local Speed Optimization only, and use CV-GX to denote AHACO with Global Speed Optimization only. For CV-GX, each ant constructs a solution with a fixed speed X (taken as 45 and 60 km/h, respectively), and after the solution is constructed, DAT60 is performed on each route in the solution. All three approaches are applied to the 30 large-scale instances of EVRP-VS with continuous speed and experimental results are presented in Fig.4. The column in Fig.4 represents the average cost of the best solution of 30 instances. ‘gap’ means the percentage deviations of the best solution obtained by each algorithm from the best of these solutions.

The results show that: 1) AHACO obtains the best solutions for 27 instances, while CV-L and CV-G60 can only find the best solutions for three instances. The average relative deviation

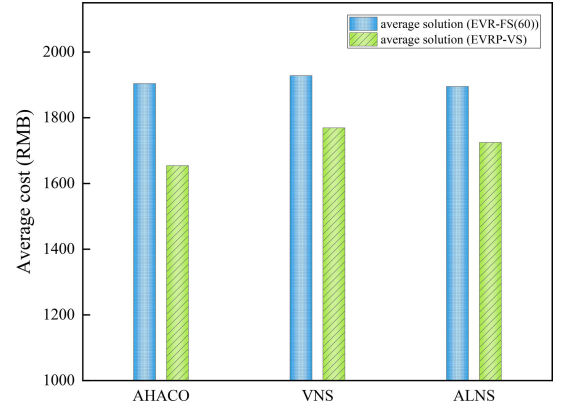


Fig. 5. Comparison of AHACO with other methods.

of the best solutions is 3.85% for CV-L, 5.53% for CV-G60, 24.7% for CV-G45, and only 0.12% for AHACO. That shows the combination of the two speed optimization strategies can significantly improve the performance of AHACO; 2) when using Local Speed Optimization alone, the deviation from the best solution is 3.85%, which is a little smaller than that of CV-G60 which uses Global Speed Optimization only. The reason may be that the Global Speed Optimization can only adjust the speed of the existing route, but cannot adjust the route of the vehicles, while the Local Speed Optimization optimizes the speed in the process of ant constructing the route, which can affect the route of the vehicles. Therefore, Local Speed Optimization plays slightly more important role than Global Speed Optimization.

F. Comparison of AHACO With Other Meta-Heuristics

Variable Neighborhood Search (VNS) [24], [30] and ALNS [3], [10] are among the most effective methods to solve VRPs [22]. AHACO is compared with them. Note that they need to be adapted to solve EVRP-VS. Specially, the initial solution of EVRP-VS is generated by using the greedy heuristic with a fixed speed of 60 km/h. The operators devised by [30] are basic ones (swap, insert, and exchange),

and our experimental results show that their performance is much worse than that of our proposed operators. Therefore, ES, 2-opt, and EM operators are used for the Shaking phase of VNS; and all of our operators are used for Variable Neighborhood Descent (VND) heuristic in VNS. All operators except Full Charge Station Removal in [10] are used in ALNS. VNS, ALNS and AHACO are used to solve EVRP-VS with continuous speed and EVRP with a fixed speed (termed EVRP-FS) of 60 km/h for 30 large-scale instances, respectively. To solve EVRP-VS, VNS and ALNS heuristics are combined with our Global Speed Optimization, termed VNS-G and ALNS-G. Once a new solution is generated by exploring a neighborhood in VNS, or by performing removal and insertion in ALNS, Global Speed Optimization is applied to improve it. Consider the average computing time (161.91s) of AHACO for large-scale instances, the runtime of each comparative algorithm (VNS, VNS-G, ALNS, and ALNS-G) is restricted as 162 seconds. Experimental results are shown in Fig.5.

The results show that, as for EVRP-FS (EVRP-VS), the average costs of the best solution of 30 instances obtained by VNS (VNS-G) and ALNS (ALNS-G) are 1928.28 (1769.32) and 1895.51 (1724.65) RMB, respectively. The performance of AHACO is better than that of VNS for both EVRP-FS and EVRP-VS. The performance of AHACO is significantly better than ALNS's for EVRP-VS, but slightly worse for EVRP-FS. For EVRP-FS, the best solutions found by AHACO are 1.35% better than those of VNS on average, but 0.19% worse than ALNS's on average. For EVRP-VS, the best solutions found by VNS-G and ALNS-G are significantly worse than those obtained by AHACO, with an average cost increase of 7.42% and 4.13%, respectively. In addition, the average solution with the fixed speed of 60 km/h (1904.1) has a higher distribution cost, i.e., 15.1% higher than that with the variable speed (1654.32) obtained by AHACO. That shows the speed optimization of ECVs can significantly decrease the cost.

VI. CONCLUSION

This paper presents a new EVRP with variable vehicle speed and soft time windows for the distribution of perishable products called EVRP-VS. It considers the energy consumption rate of ECVs with refrigeration, which has a nonlinear relationship with travel speed and a linear one with weight. ECVs can actively adjust the travel speed to improve the driving range and decrease distribution cost. The perishable products must be cooled during the distribution by using refrigerated ECVs. ECVs can be charged at public recharging stations along their routes. A model of EVRP-VS is formulated. A combined objective function is devised, which considers the number of vehicles, total energy consumption, and total customer delay time.

We develop a hybrid ACO with local search and adaptive operator selection mode to solve EVRP-VS. A two-stage speed optimization strategy is proposed to optimize the speed of ECVs. Based on experimental comparisons, we conclude that 1) AHACO can effectively solve EVRP-VS in comparison with CPLEX, VNS, and ALNS. CPLEX can find optimal solutions for problem instances of EVRP-VS with discretized

speed with 5 customers. AHACO can find high-quality solutions for instances with up to 100 customers; 2) with ASM, AHACO can dynamically adjust the selection probability of each operator according to its previous performance, and has better performance than SHACO with SSM; 3) LS operators can greatly improve the performance of ACO. MD, MS, and WS operators have stronger ability to improve the solution than ES, 2-opt and EM operators; and 4) the speed optimization of ECVs can dramatically decrease the distribution cost. The two-stage speed optimization strategy can significantly improve the performance of AHACO and brings lower cost than AHACO with Local or Global Speed Optimization only.

Our future work intends to study EVRP using ECVs with multi-compartment for multiple product segments with different temperature zones, and develop solution approaches [31], [32] for large-scale EVRP-VS.

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